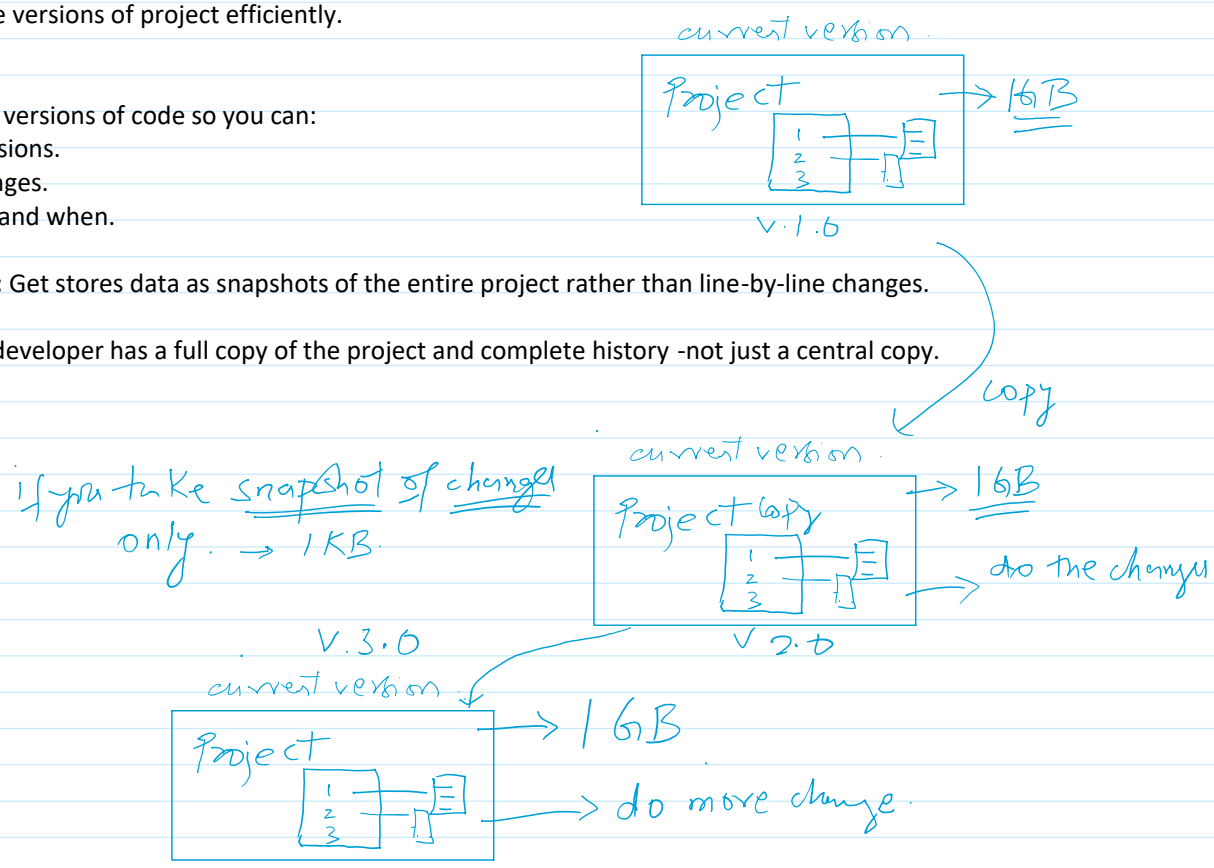
# GIT

21 June 2025 20:26

**What is Git?**
Git is a Distributed Version Control System (DVCS) that allows developer to track changes in their code, collaborate with others and manage multiple versions of project efficiently.

**What is version control?**
Get helps manage different versions of code so you can:

- Go back to previous versions.
- View the history of changes.
- See who changed what and when.




**Snapshots, not differences:** Get stores data as snapshots of the entire project rather than line-by-line changes.

**Distributed System:** Every developer has a full copy of the project and complete history -not just a central copy.

**Important Git Terminologies**

| Term | Description |
|---|---|
| **Repository(Repo)** | A folder containing all project files and their history cracked by Git |
| **Commit** | A snapshot of changes with a unique id. |
| **Branch** | A parallel version of the Repository to work on features without affecting main code. |
| **Merge** | Combine changes from one branch into another. |
| **Clone** | Makes a local copy of a remote repository. |
| **Pull** | Fetch and integrate changes from the remote repository. |
| **Push** | Upload your local changes to the remote repository. |

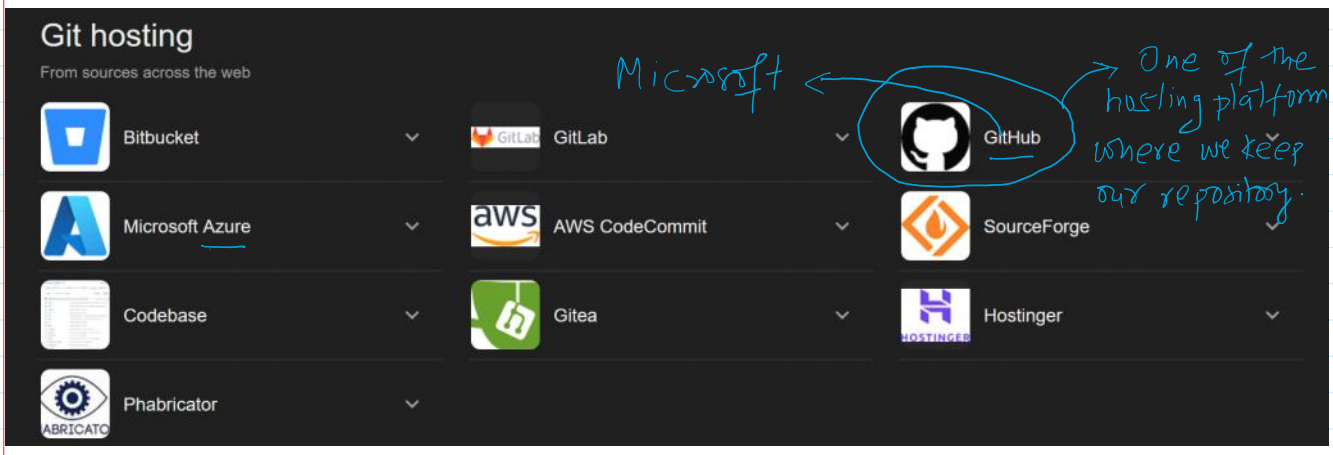
**Basic Git Commands**

| Command | Description |
|---|---|
| **git init** | Initialize a new get repository. |
| **git status** | Shows current changes and file status. |
| **git add <file>** | Stage changes for commit. |
| **git commit -m "message"** | Save the changes to history with a message. |
| **git log** | View the history of commits. |

| | |
|---|---|
| **git branch** | List or create branches. |
| **git checkout<branch>** | Switch to a branch. |
| **Git merge <branch>** | Merge another branch into the current one. |

**What is GitHub?**

**GitHub** is a cloud based hosting platform for git repository. It allows developers to store their code online, collaborate on projects and manage issues will request and contributions.




**Who developed Git?**

**Git is developed by Linus Trovaldus who is also the developer of Linux kernel.**

**Features of GitHub**

1. **Remote Repository: Share and collaborate on get projects with other across the globe.**
2. **Pull Request (PR):** Summit your changes for review before merging them into the main code base.
3. **Issues:** Track bugs, task, or feature request in a structured way.
4. **Forks:** Make a personal copy of someone else's repository to work on.
5. **Actions:** Automate workflows like testing and development.

The primary command used to generate SSH keys for used with Git **ssh-keygen** command. This command is part of the OpenSSH suit, which is typically installed by default on Linux and MacOS, and is available through Git Bash or Windows Subsystem for Linux(WSL) on Windows.

Comment.

ssh-keygen -t <key_type> -b <key_length> -C "<comments>"

number of bits in the key → 2048 — " " "
→ 4096 — bit rsa key
→ 8192 — " " "

Specifies the type of key to generate.

Common types include:

ed25519 : A modern, highly secure, and efficient algorithm.

rsa: A widely supported algorithm. When using rsa, it's recomended to specify key-length.